# CPE 301: Embedded Systems Design Final Project Part 3 Assignment

**Group members' names: Noah Clark, Luis Angel Sarellano, and Leonardo Ramirez Jimenez**

## 1. Project Motivation/Introduction

Energy waste and overheated equipment are very common issues in hot environments or indoors, such as workshops, offices, garages, and rooms. Heating and cooling systems are typically used without checking an area's conditions, which could possibly lead to unnecessary energy waste. Motors and other electronic equipment also face similar problems in higher temperature conditions, increasing possible risks of reduced efficiency or failures that could lead to safety concerns. This Overheating Monitoring System is designed not only to improve energy efficiency but also safety. The system will continuously check the temperature in spaces and give alerts when conditions are no longer considered safe or efficient. This system is useful for many people. That includes homeowners, workshop workers without access to advanced monitoring systems, homeowners, and even students. An embedded system is a good solution since it can continuously run and respond to real-time situations while using energy efficiently.

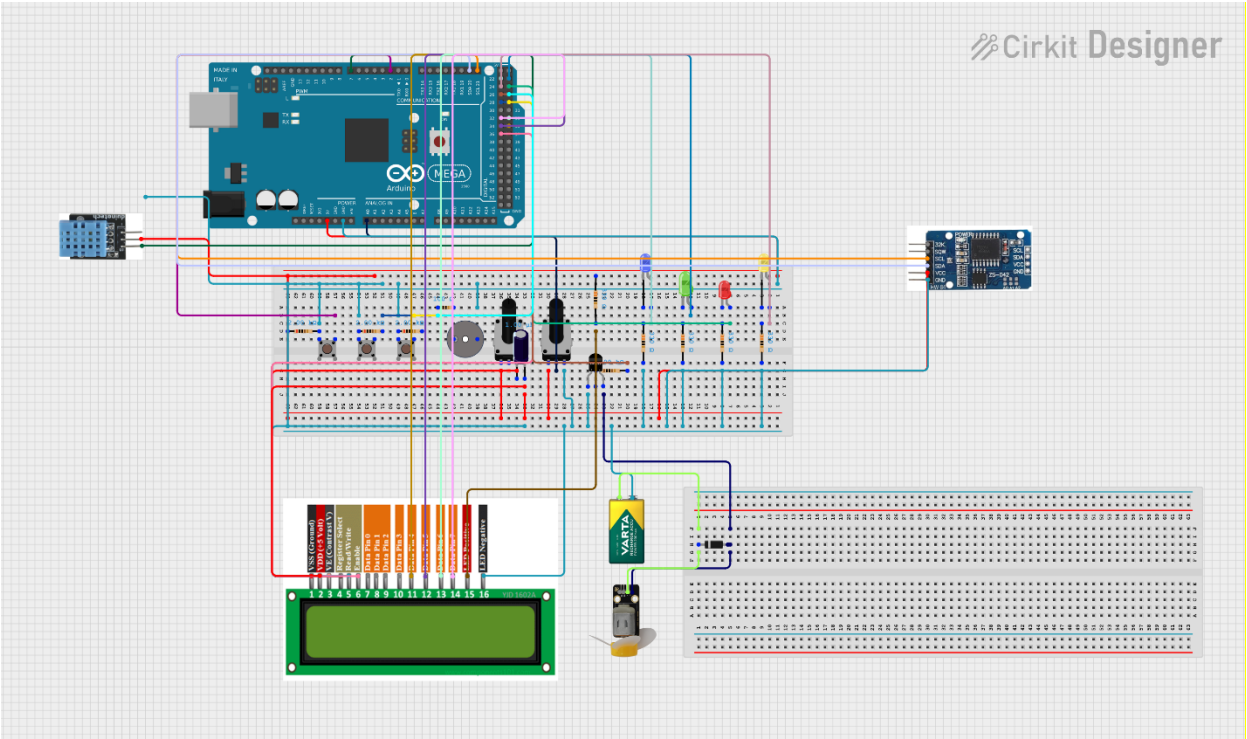
## 2. Proposed System

The system will use a DHT11 to measure temperature, a real-time clock module that will track and log warnings, as well as an LCD (16X2) display.

The main features should include continuous checking of temperature, defined thresholds, audio and visual alarms that warn during critical conditions, a real-time clock display, and a user configuration.

The system will look at measured values and compare them to preset safety and efficiency limits. If the temperature exceeds the range that was set, the LCD will display a suggestion to improve ventilation. There will also be LED indicators and an alarm alerting if the threshold exceeds the max. Users will be able to interact with the system using potentiometers and push buttons to adjust temperature levels. LEDs will provide quick feedback, while the sound will provide immediate warning when needed. In all, the system should react to the environment and promote energy efficiency as well as prevent overheated equipment.

## 2.1 System Schematic Diagram




## 2.2 Component List

- 3 Push Buttons
- Buzzer: The buzzer was learned how to set up and use back in Lab 5. To set the correct frequency for the buzzer, the lecture “Timer Theory” was utilized.
- 2 Potentiometers: The 2 potentiometers were used to learn how to set up the LCD and how to use them in Lab 7.
- Blue LED
- Green LED
- Yellow LED
- Red LED
- 5 330 ohms Resistors
- 3 2k ohms Resistors
- 1k ohms Resistors
- 100 ohms Resistors
- 9V Battery
- LCD Display: The LCD Display was learned how to use and how to set up with the potentiometers back in Lab 7. The lecture, “Arduino Display 2”, was also used to learn more about the LCD and how it works.

- Fan
- 2 Breadboards
- Transistor
- Diode: Diodes were learned on how to use and how they work during the first lecture, “Circuits Review”.
- Capacitor: Capacitors were learned on how to use and how they work during the first lecture, “Circuits Review”.
- DHT11
- RTC: The Real-Time Clock was learned during the lecture, “RTC, Attach Interrupt, Milli”. This helps to learn how the RTC is connected to the Arduino and what it does.
- ArduinoMega2560: The datasheet and pin configurations were learned during the “GPIO Theory” lecture.
- Wires

## 3. System Behavior

### Power-On Behavior

When our system is powered on, the Arduino initializes all sensors and output devices, including the DHT11 sensor, RTC module, 16x2 LCD Display, LEDs, buzzer, potentiometers, and a fan. During startup, the LCD displays a cute welcome message, the RTC synchronizes and begins tracking time, and default temperature thresholds are either manually input or loaded in. The system then enters monitoring mode. The LCD will display the current temperature and current time.

### Normal Operating Behavior

During normal operation, our system continuously samples data using the DHT11 sensor at a set interval. The Arduino will read temperature values, compare those readings to the predefined threshold values, update the LCD in real time, and maintain RTC time tracking for event logging. If temperature values remain under our threshold values, then the green LED remains ON, the fan remains OFF, no alarm is triggered, and the LCD remains the same.

### Warning Condition Behavior

If the temperature exceeds the defined warning threshold and is still below the critical level, the yellow LED turns on, the fan is activated to increase ventilation, and the LCD will display on the first line “ACTIVE FAN ON” and the RTC will log when this warning occurs. The fan and LED will remain active until the temperature falls below the threshold.

### Critical Behavior

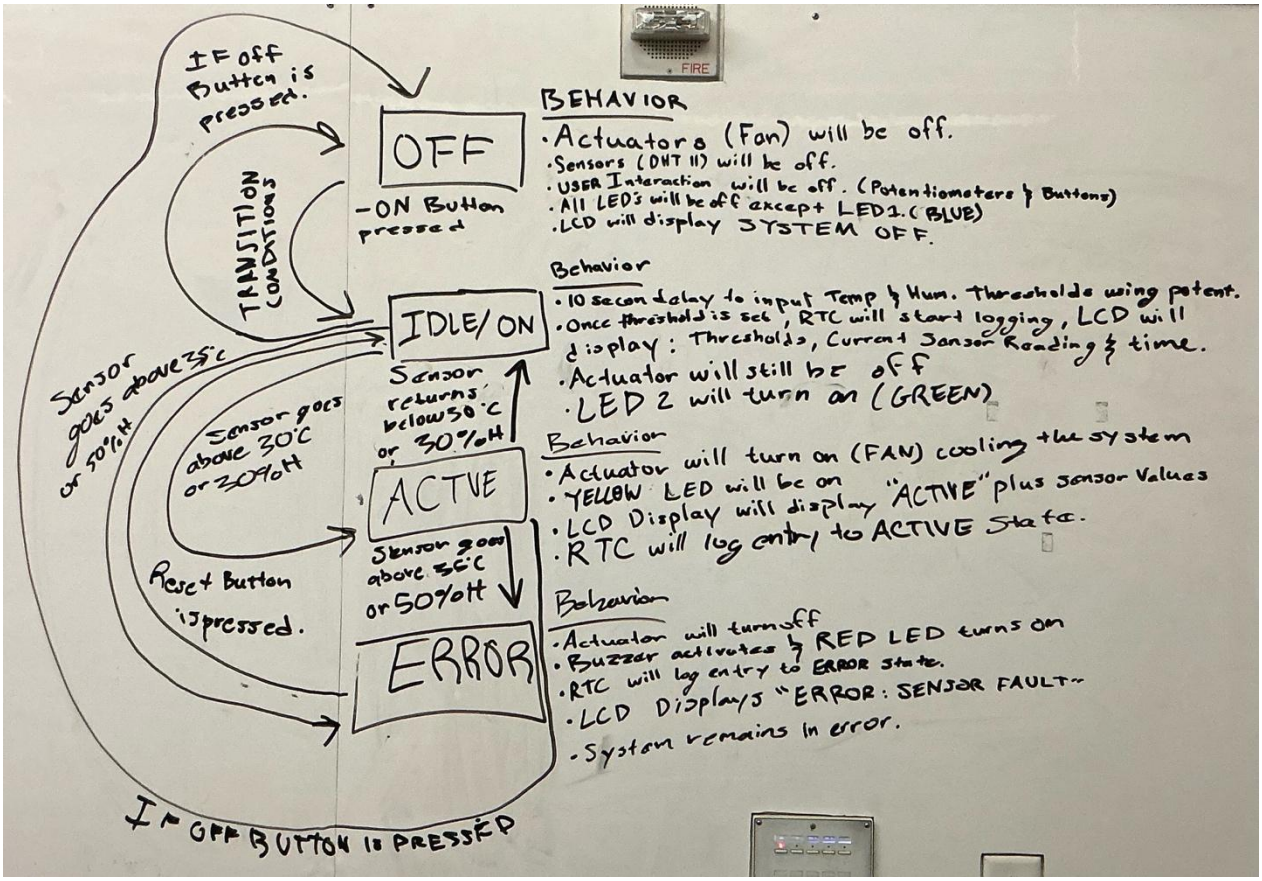
If the temperature exceeds their maximum allowable threshold, the red LED will turn on, the buzzer will activate continuously, and the LCD will display “ERROR OVERHEAT”. Even if the temperature drops back down to allowable values, the system will not work until the user

manually resets the system. While the system is off, all monitoring has stopped, the fan is deactivated, and all LEDs and buzzers turn OFF, but the RTC still keeps track of time.

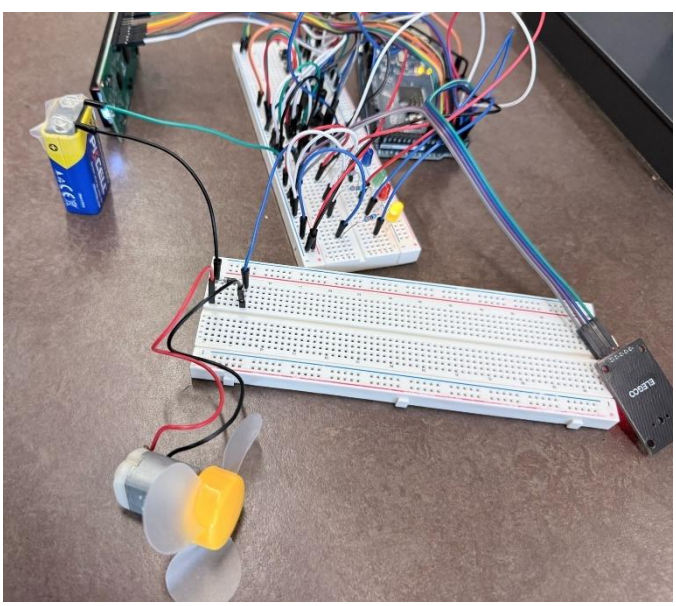
## 3.1 Module Design

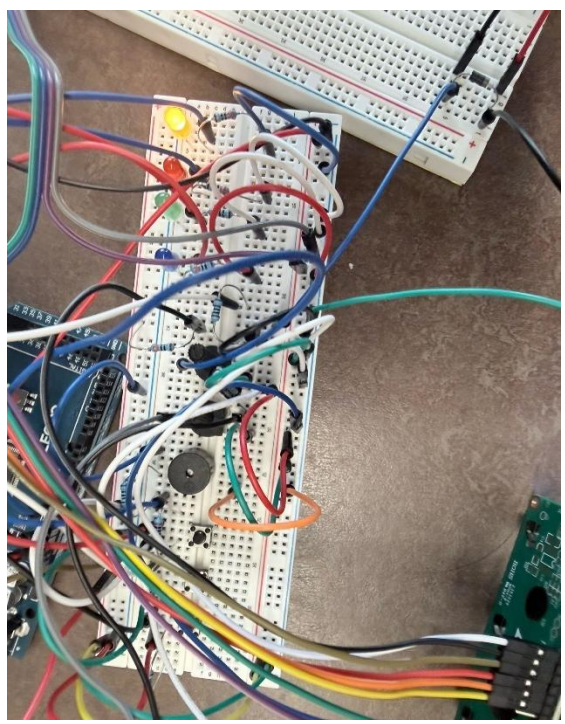
- Sensors: The system will use a DHT11 to track temperature.
- User Interface: Feedback of the states of the system will be provided with the LCD display, which controls everything, the LED indicators, and a buzzer.
- Actuator: A fan, when triggered, will ventilate the system during warning states.
- User interaction: The on and off button, the potentiometer will give our thresholds for our temperature.

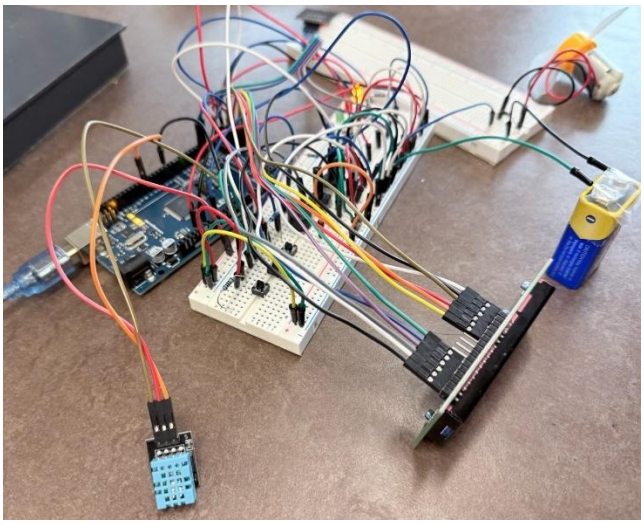
## 3.2 Control Logic (State Diagram)

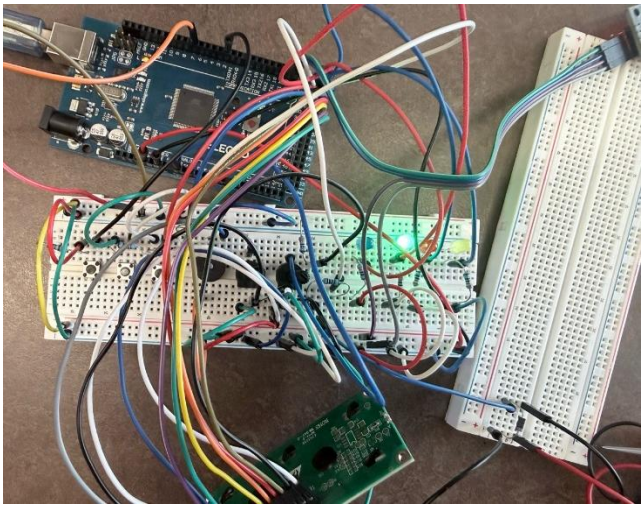


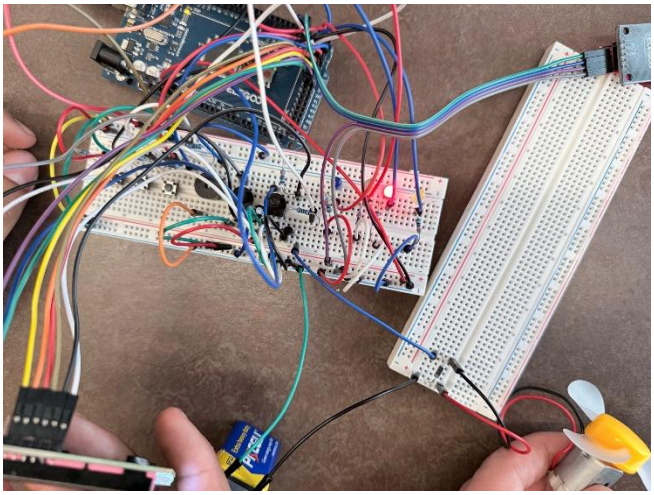

## 4. Result:

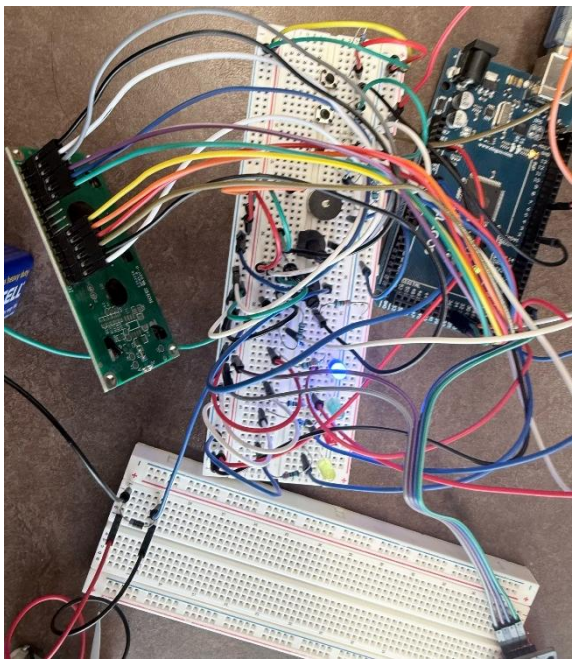


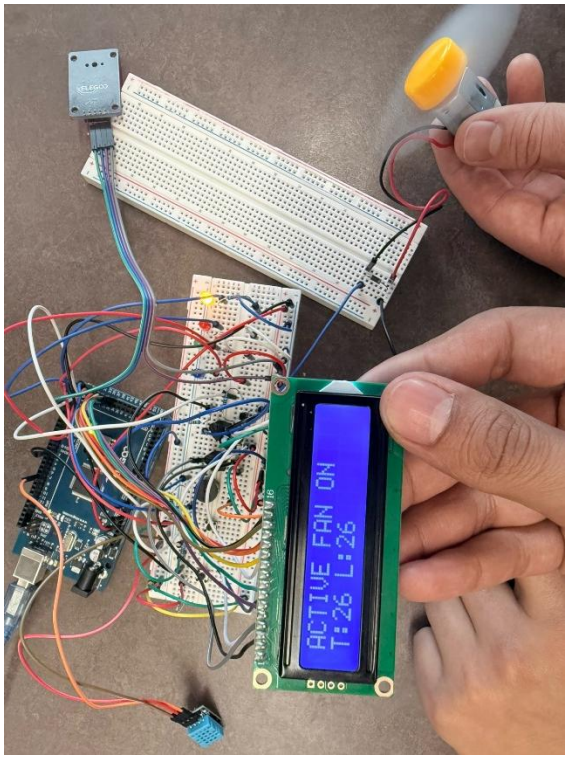











ACTIVE FAN ON
T:26 L:26

```
Serial Monitor ×   Output
Message (Enter to send message to 'Arduino Mega or Mega 2560' on 'COM3')
[11:16:43] ENTERED IDLE STATE | Temp=27C | Warning=29C | Max=32C
[11:16:43] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:45] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:47] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:49] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:51] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:53] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:55] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:57] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
[11:16:59] IDLE DATA LOG | Temp=27C | Warning=29C | Max=32C
```

```
[11:31:40] ENTERED ACTIVE STATE | Temp=30C | Warning=29C | Max=32C
[11:31:40] ACTIVE DATA LOG | Temp=30C | Warning=29C | Max=32C
[11:31:42] ACTIVE DATA LOG | Temp=30C | Warning=29C | Max=32C
[11:31:44] ACTIVE DATA LOG | Temp=30C | Warning=29C | Max=32C
[11:31:46] ACTIVE DATA LOG | Temp=30C | Warning=29C | Max=32C
[11:31:48] ACTIVE DATA LOG | Temp=29C | Warning=29C | Max=32C
[11:31:50] ACTIVE DATA LOG | Temp=29C | Warning=29C | Max=32C
[11:31:52] ACTIVE DATA LOG | Temp=29C | Warning=29C | Max=32C
[11:31:54] ACTIVE DATA LOG | Temp=29C | Warning=29C | Max=32C
[11:31:56] ACTIVE DATA LOG | Temp=29C | Warning=29C | Max=32C
```

```
[11:40:16] ENTERED ERROR STATE | Temp=37C | Warning=29C | Max=32C
[11:40:16] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:18] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:20] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:22] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:24] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:26] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:28] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:30] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
[11:40:32] ERROR STILL ACTIVE | Temp=37C | Warning=29C | Max=32C
```

## 5. Environmental & Energy Impact

### Energy Efficiency:

The system will minimize energy consumption by telling the user real-time data that the motors and other electronic equipment are overheating and automatically powering down the motor or equipment. This can also help reduce waste because once the alert is on, the user can fix the problem without having to replace the equipment or motor.

### Design Safety:

Since the system will have mostly sensors, a screen, and a real-time clock module, resistors can make the right amount of voltage and current flow through the system to ensure its own safety.

### Affordability:

The system will have components with a cost that is neither too high nor too low. The temperature sensor and the LCD screen will likely be the most expensive of all the other components. Also, we will be using great resistors that won't break if the circuit gets damaged.

### Sustainability:

The system will have to be covered in a material that not only can change the temperature in its surroundings, but also in a material that can protect the system from being destroyed easily, so the system doesn't have to be replaced constantly.

### Accessibility:

In this system, the user only needs to turn the system on and off. When the system is turned on, it will automatically sense the temperature in the room, and once it does, the system will show the user on the LCD screen what the temperature is surrounding the user's equipment.

## 6. Conclusion

In all, we had a successful system. We were able to have it fully operate correctly; the fan would turn on to cool our system if it got too hot, or succeeded the threshold set, we would get warnings, and the system would shut down if overheated. Some limitations would be that our system has a lot of cords and it might be a little hard to navigate through. But if we had a casing it could probably look a lot better. Overall, we were able to complete what we wanted.